Fuzzing findings

Overview

Definition of "break":

A break occurs when the application cannot be built or execution stops due to an unhandled exception during the build/import process in Visual Studio.

Failure Types Observed:

- Build-time crashes (critical)
- Silent fallback to defaults in cases of invalid, blank or random inputs
- Silent data loss (sections/questions not imported)

Syntax-level Fuzzing:

Special characters such as '>' at the beginning of a line cause global YAML parsing failures and prevent the application from building.

Overall Result:

The system is highly fragile against malformed configuration input.
Many invalid inputs lead to build-time crashes instead of graceful
error handling or input validation.

/Documentation

config.yaml

- *CourseName*
  - Breaks when left empty
  - Breaks when the symbol “<“ is used; all other keyboard symbols are accepted
- *Summary*
  - can be left empty
  - Allows random inputs without breaking
- *Visible*
  - Defaults to false when left blank
  - Breaks on random inputs
- *CategoryId*
  - Can be left empty
  - Breaks on random inputs
- *Category*
  - Can be left empty
  - Accepts random inputs without breaking
- *StartDate & EndDate*
  - Cannot be left empty
  - DateTime format must remain intact
  - Breaks on random inputs
- *IdNumber*

    - Can be left empty
    - Only numeric values are allowed; non-numeric input causes a break

- *ForcedLanguage*
    - Can be left empty; no language is forced in this case
    - Other languages must be installed in Moodle beforehand, otherwise the system falls back to the default language defined in
      Site administration → General → Language settings
    - Random input reverts the language to the default value
- *ShowGrades*
    - Defaults to false when left empty
    - Breaks on random inputs
- *NumberOfAnnouncements*
    - Defaults to 0 if left empty, below 0, or above 10
    - Only numeric values are allowed; non-numeric input causes a break
- *ShowActivityDates*
    - Defaults to false when left empty
    - Breaks on random inputs

- *ShowCompletionConditions*
    - Defaults to false when left empty
    - Breaks on random inputs
- *EnableCompletionTracking* (Code should be moved up by one line in config.yaml, since its above ShowCompletionConditions in moodle)
    - Defaults to false when left empty
    - Breaks on random inputs

- *GroupMode*
    - Defaults to 0 (No groups) when left empty
    - Values below 0 or above 1 are ignored
    - The value 2 (Visible groups) is ignored and defaults to 0, indicating a mismatch between Moodle configuration options and the importer implementation
    - Only numeric values are allowed; non-numeric input causes a break
- *ForceGroupMode*
    - Defaults to false when left empty
    - Random inputs are not allowed and cause a break
- *QuestionListPaths*
    - Listing the same question path multiple times does not cause errors and does not lead to duplicated questions in Moodle
    - Breaks if:
        - Not all questions in the referenced question folder are listed
        - Paths contain typos or are incorrect
        - Listed paths or files do not exist

/Documentation/Sections

sections.yaml

- Sections
    - The system remains functional if an existing section is not listed; however, the omitted section will not appear in the imported Moodle course
    - Copying and pasting the same section path multiple times results in multiple instances of that section (e.g., listing a quiz section five times causes it to appear five times)
    - If a section path contains a typo, the section is not displayed and no warning is shown
    - Using - SectionPath: with random input does not break the build; the course is imported successfully, but the random input does not add or modify any sections

/Documentation/Questions

General

- *QuestionName*
    - No issues when left blank
    - Accepts every input without breaking
- *Category*
    - No issues when left blank
    - Accepts every input without breaking
- *QuestionId*
    - Breaks when left blank
    - Only numeric values are allowed
- *ShuffleAnswers*
    - *Has no effect, even when set to true*
- *QuestionText*
    - No issues when left blank
    - Accepts every input without breaking
- *QuestionTextFormat*
    - Does not allow fuzzing, breaks otherwise
    - Defaults to HTML when left empty or removed
- *QuestionType*
    - Breaks on random inputs
    - Defaults to SINGLECHOICE when left empty
- *DefaultMark*
    - Questions are not graded when left empty
    - Values below -1 prevent the quiz from being started and cause an error
- *ClearIncorrectResponses*
    - Breaks on random inputs
    - Defaults to false when left empty
- *ShowNumberOfCorrectAnswers*
    - Breaks on random inputs

    - Defaults to false when left empty
- *AnswerText*
    - *Can be left blank without issues*
    - *Does allow random inputs*
- *AnswerTextFormat*
    - *Defaults to HTML when left blank*
    - *Doesn’t allow random inputs*